Theory and manual

for

AlkCalc

Simon Euchner*

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* Electronic Mail: ⟨euchner.se@gmail.com⟩

Contents

I. Introduction **1**

II. Theory **1**

III. Manual **9**

References **16**

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I. Introduction

Bound single-electron eigenfunctions of alkali-metal atoms and alkaline-earth metal ions are of relevance whenever properties of Rydberg atoms are of interest (e.g. in Quantum Computing and Information, Quantum Simulation, pure Rydberg Physics, Quantum Optics, ...). Generally, because except for the Hydrogen atom the time-independent Schrödinger equation cannot be solved analytically, there is the need to obtain the eigenenergies (atomic spectrum) and eigenstates numerically. The library `AlkCalc` is intended for this purpose and information is given in the various provided README files. However, there is no discussion on the theory behind the code.

This document's purpose is to define the Hamiltonian which `AlkCalc` is designed to diagonalise. Moreover, it formally defines the numerically computed quantities and hence serves as a reference for the user. In particular, the final section is a reference manual or the library functions.

II. Theory

The starting point is an alkali-metal atom or alkaline-earth-metal ion. We are interested in systems where all electrons are in closed shells except for one (e.g., ^{1}H, ^{38}Rb, ^{88}Sr^{+}). This allows a Kepler-problem description for the 'effective nucleus' (the nucleus 'screened' by the closed-shell electrons) and valence electron. To be concrete, we study the Hamiltonian

$$H = \frac{\boldsymbol{p}^2}{2m} + V(\|\boldsymbol{r}\|\,; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2)\,, \tag{II.1}$$

where $m = Mm_{\mathrm{e}}/(M + m_{\mathrm{e}})$ is the reduced mass of the system, composed of the electron's mass m_{e} and the atom's or ion's mass M. The centre-of-mass motion is decoupled, such that $\boldsymbol{r}, \boldsymbol{p}$ are the relative canonical co-ordinates, i.e., $\boldsymbol{r}$ is the distance vector between the centre-of-mass position of the effective nucleus and the valence electron. Moreover, $\boldsymbol{L}$, $\boldsymbol{S}$, and $\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$ are the orbital angular momentum operator, the electron's spin operator, and the total angular momentum operator, respectively. The potential V is not a simple Coulomb potential, but a generalised potential which includes different contributions (these arise when one starts with the full many-body electron Hamiltonian and incorporates relativistic and quantum electrodynamics effects). Following Refs. [1–3] we write:

$$V(\|\boldsymbol{r}\|\,; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2) = V_{\mathrm{C}}(\boldsymbol{r}) + V_{\mathrm{P}}(\boldsymbol{r}) + V_{\mathrm{R}}(\boldsymbol{r}; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2)\,. \tag{II.2}$$

The first contribution is

$$V_{\mathrm{C}}(r) = -\frac{e^2}{4\pi\varepsilon_0}\frac{Z_{\mathrm{n}}(r)}{r}\,,\ Z_{\mathrm{n}}(r) = Z_{\mathrm{c}} + (Z - Z_{\mathrm{c}})\mathrm{e}^{-k_1 r} + k_2 r\mathrm{e}^{-k_3 r} - k_4 r^2 \mathrm{e}^{-k_3 r}\,. \tag{II.3}$$

This is a modified Coulomb potential, adjusted for small radii due to the effective nucleus. Here, Z is the nuclear charge, Z_{c} the nuclear charge of the effective nucleus, and the k_j ($j = 1, 2, 3, 4$) are empirically determined fitting parameters; see Refs. [1–3]. The second contribution is

$$V_{\mathrm{P}}(r) = -\frac{e^2}{(4\pi\varepsilon_0)^2}\frac{\alpha_{\mathrm{d}}}{2r^4}\left[1 - \exp\left(-\frac{r^6}{r_{\mathrm{c}}^6}\right)\right], \tag{II.4}$$

which accounts for the polarization of the screened nucleus due to the presence of the valence electron. Here, r_c is the effective nucleus' size and α_d its polariseability. Finally,

$$V_R(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2) = \frac{1}{2m^2c^2}\frac{V'_{NR}(r)}{rN(r)}\boldsymbol{L}\cdot\boldsymbol{S}\,,\ N(r) = \left[1 - \frac{V_{NR}(r)}{2mc^2}\right]^2,\ V_{NR}(r) = V_C(r) + V_P(r)\,, \tag{II.5}$$

which describes (relativistic) coupling between the orbital angular momentum and the valence electrons' spin. In this expression, V_{NR} is the non-relativistic model potential and c the speed of light. In Refs. [1, 3] this potential is used for ions. We note that in Ref. [4] the simplified version

$$V_R(r) = \frac{\alpha}{2r^3}\boldsymbol{L}\cdot\boldsymbol{S} \tag{II.6}$$

is employed for atoms. Here, α is the fine-structure constant. However, we use the first, more precise, version for both ions and atoms. The spin-operator $\boldsymbol{S}$ in Eqs. (II.6, II.5) is conveniently expressed as

$$\boldsymbol{S} = \frac{\hbar}{2}\begin{bmatrix}\sigma_x\\ \sigma_y\\ \sigma_z\end{bmatrix}, \tag{II.7}$$

with the Pauli matrices

$$\sigma_x = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix},\ \sigma_y = \begin{bmatrix}0 & -\mathrm{i}\\ \mathrm{i} & 0\end{bmatrix},\ \sigma_z = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}, \tag{II.8}$$

which is possible because we deal with a single electron, i.e., a particle with spin 1/2. Moreover, we rewrite the product $\boldsymbol{L}\cdot\boldsymbol{S}$ in terms of the total angular momentum operator $\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$:

$$\boldsymbol{L}\cdot\boldsymbol{S} = \frac{1}{2}\left[\boldsymbol{J}^2 - \boldsymbol{L}^2 - \boldsymbol{S}^2\right]. \tag{II.9}$$

For further simplification the square of the momentum operator is expressed in spherical coordinates,

$$\boldsymbol{p}^2 = (-\mathrm{i}\hbar\nabla)^2 = -\hbar^2\nabla^2 = -\frac{\hbar^2}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\boldsymbol{L}^2}{r^2}. \tag{II.10}$$

The second contribution is the standard centrifugal energy contribution, analogously to the Kepler problem. With these simplification we rewrite the Hamiltonian as

$$H = -\frac{\hbar^2}{2mr^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + V_{\mathrm{eff}}(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2)\,,\ V_{\mathrm{eff}}(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2) = \frac{\boldsymbol{L}^2}{2mr^2} + V(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2)\,. \tag{II.11}$$

The full Hilbert space this Hamiltonian acts on is split into an angular momentum factor and a radial one. By construction, a basis of the angular momentum Hilbert space is given by the states $|l, s, j, m_j\rangle$, which are eigenstates of $\boldsymbol{L}^2$, $\boldsymbol{S}^2$, $\boldsymbol{J}^2$, and J_z, i.e., constitute the coupled basis, also called 'fine-structure basis'. Note that it must be possible to construct an eigenbasis of $\boldsymbol{L}^2$, $\boldsymbol{S}^2$, $\boldsymbol{J}^2$, and J_z simultaneously, because they commute pairwise. In particular, these states are constructed as linear combinations of the fine-structure basis states $Y_{l,m_l} \otimes \boldsymbol{\chi}_{s,m_s}$, employing Clebsch-Gordan coefficients. Here, $Y_{l,m_l} : [0,\pi] \times [0, 2\pi] \to \mathbb{C}$ are spherical harmonics defined as (the so-called

'Condon-Shortley phase' we include in the definition of the Legendre polynomials below, i.e., the factor $(-1)^{m_l}$ is *not* missing here; see e.g. Ref. [5])

$$Y_{l,m_l}(\vartheta,\varphi) = \sqrt{\frac{2l+1}{4\pi}\frac{(l-m_l)!}{(l+m_l)!}}P_l^{m_l}(\cos(\vartheta))\,\mathrm{e}^{\mathrm{i}m_l\varphi}\,, \tag{II.12}$$

where the associated Legendre polynomials $P_l^{m_l} : [-1,1] \to \mathbb{R}$ are defined for $l = 0,1,2,\ldots$ and $m_l = 0,1,\ldots,l-1,l$, via the recursion relation

$$(l-m_l)P_l^{m_l}(x) = x(2l-1)P_{l-1}^{m_l}(x) - (l+m_l-1)P_{l-2}^{m_l}(x)\,, \tag{II.13}$$

with starting point

$$P_{m_l}^{m_l}(x) = (-1)^{m_l}\frac{(2m_l)!}{2^{m_l}m_l!}(1-x^2)^{m_l/2}\,, \tag{II.14}$$

in combination with $P_l^{m_l}(x) = 0$ for $l < m_j$. Negative values of m_l are treated with the relation

$$P_l^{-m_l} = (-1)^{m_l}\frac{(l-m_l)!}{(l+m_l)!}P_l^{m_l}\,. \tag{II.15}$$

Moreover, $\boldsymbol{\chi}_{s,m_s}$ are two-spinors defined as

$$\boldsymbol{\chi}_{\frac{1}{2},\frac{1}{2}} = \begin{bmatrix}1\\0\end{bmatrix},\ \boldsymbol{\chi}_{\frac{1}{2},-\frac{1}{2}} = \begin{bmatrix}0\\1\end{bmatrix}. \tag{II.16}$$

Explicitly, (for the case of interest $s = 1/2$), we find the following expressions for the coupled basis states $\boldsymbol{\Phi}_{l,s,j,m_j}$:

$$\boldsymbol{\Phi}_{l,s,j,m_j} = \langle Y_{l,m_j-s}\otimes\boldsymbol{\chi}_{s,s},\boldsymbol{\Phi}_{l,s,j,m_j}\rangle Y_{l,m_j-s}\otimes\boldsymbol{\chi}_{s,s} + \langle Y_{l,m_j+s}\otimes\boldsymbol{\chi}_{s,-s},\boldsymbol{\Phi}_{l,s,j,m_j}\rangle Y_{l,m_j+s}\otimes\boldsymbol{\chi}_{s,-s}\,, \tag{II.17}$$

where the scalar products are (possibly vanishing) Clebsch-Gordan coefficients. We further find that these states transform according to the irreducible representations of the group SU(2) under the action of the SU(2)-representation $\rho(\boldsymbol{\alpha}) = \exp(\mathrm{i}\boldsymbol{\alpha}\cdot\boldsymbol{J})$, where $\boldsymbol{\alpha}\in\mathbb{R}^3$ is a vector of length $\|\boldsymbol{\alpha}\|\in[0,2\pi)$. Moreover, the angular momentum Hamiltonian is invariant under the action of ρ, such that it is automatically block diagonal. The irreducible representations of SU(2) are $j(j+1)$-dimensional, with $j = 0,1/2,1,3/2,2,\ldots$, such that the blocks have dimensions $\mu_j j(j+1)$, where μ_j is the multiplicity of the irreducible representation j in the direct sum decomposition of ρ. The eigenvalues associated to the blocks are operators on the radial Hilbert space. The multiplicities are $\mu_j = 2$ and $\mu_0 = 0$, because for each fixed j there are two possibilities: $l = j-1/2$ and $l = j+1/2$. Explicitly, the Hamiltonian is written as

$$\bigoplus_{m_j=-j}^{j}\langle\boldsymbol{\Phi}_{l,s,j,m_j},H\boldsymbol{\Phi}_{l,s,j,m_j}\rangle \equiv \bigoplus_{m_j=-j}^{j}H_{l,s,j}\,, \tag{II.18}$$

where (with the arbitrary choice $m_j = j$) we defined

$$H_{l,s,j} = \langle\boldsymbol{\Phi}_{l,s,j,j},H\boldsymbol{\Phi}_{l,s,j,j}\rangle = -\frac{\hbar^2}{2mr^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + V_{\mathrm{eff}}(r;\hbar^2 l[l+1],\hbar^2 s[s+1],\hbar^2 j[j+1])\,. \tag{II.19}$$

We note that the fact that the matrix element does not depend on m_j is a consequence of the SU(2)-symmetry of the Hamiltonian. Therefore, exactly this is the reason for the energies being degenerate in the total magnetic angular momentum quantum number m_j[1] The symmetry consideration up until now simplified the problem to the block associated to the irreducible representation of SU(2). What is left to solve are the eigenproblems

$$H_{l,s,j} R_{n,l,s,j} = E_{n,l,s,j} R_{n,l,s,j} \,, \tag{II.20}$$

where $R_{n,l,s}$ is the n-th (radial) eigenstate of $H_{l,s,j}$. Therefore, if we manage to obtain $R_{n,l,s}$ the full spectrum is known. Of course, resonances (states associated to the channel Φ_{l,s,j,m_j} which are not bound) exist, but in the following we will focus on bound states.

This is now the part where `AlkCalc` comes into play: the radial eigenstates cannot be obtained analytically. `AlkCalc` provides a platform to solve for these states numerically. To make the problem addressable by a computer, we express the eigenproblem in terms of dimensionless quantities. To this end we express the Hamiltonians $H_{l,s,j}$ in units of Hartree, $E_\mathrm{H} = 1/4\pi\varepsilon_0 a_\mathrm{B}$, where $a_\mathrm{B} = \hbar/(m_\mathrm{e} c\alpha) \approx 52.9\,\mathrm{pm}$ (m_e: electron mass) is Bohr's radius. The radial eigenfunctions we parametrise as

$$R_{n,l,s,j}(r) = \frac{1}{\sqrt{a_\mathrm{B}^3}} \frac{f_{n,l,s,j}(r/a_\mathrm{B})}{r/a_\mathrm{B}} \,. \tag{II.21}$$

where $f_{n,l,s,j}$ is dimensionless. The eigenproblem in Eq. (II.20) in the new eigenfunctions takes the form

$$\left[-\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}t^2} + \frac{l(l+1)}{2t^2} + C\frac{V(a_\mathrm{B} t; \hbar^2 l[l+1], \hbar^2 s[s+1], \hbar^2 j[j+1])}{E_\mathrm{H}} \right] f_{n,l,s,j}(t) = \bar{E}_{n,l,s,j} f_{n,l,s,j}(t) \,, \tag{II.22}$$

where we defined

$$C = \frac{1}{1 + m_\mathrm{e}/M} \tag{II.23}$$

and the eigenenergies may be extracted from the relation

$$\frac{E_{l,s,j,n}}{E_\mathrm{H}} = \frac{\bar{E}_{n,l,s,j}}{C} \,. \tag{II.24}$$

Rewriting the normalisation condition in terms of the eigenfunctions $f_{n,l,s,j}$ leads to

$$1 = \int_0^\infty \mathrm{d}r\, r^2 R_{n,l,s,j}^2(r) = \int_0^\infty (a_\mathrm{B}\,\mathrm{d}t)(a_\mathrm{B} t)^2 \frac{1}{a_\mathrm{B}^3} \frac{f_{n,l,s,j}^2(t)}{t^2} = \int_0^\infty \mathrm{d}t\, f_{n,l,s,j}^2(t) \,. \tag{II.25}$$

Here we assumed that the (radial) eigenfunctions are real-valued, which is justified as the Hamiltonian is not only an hermitian but also a symmetric operator. In conclusion, computing the

1 Generally, there is no reason that the blocks $H_{l,s,j}$ are separate blocks for $l = j + 1/2$ and $l = j - 1/2$, because they correspond to the same irreducible representation j. However, the states we choose are already chosen in a ways such that the blocks are already separated. In this case we know how we should choose the states already from the fact that the Hamiltonian explicitly depends on $\boldsymbol{L}$: if it would not be so obvious, we would not immediately know what the multiplicity index l (and s) should physically be to sort the eigenvalues into separate blocks.

normalised eigenfunctions $f_{n,l,s,j}$ yields the (automatically normalised) radial eigenstates by employing Eq. (II.21) for the conversion.

To solve the eigenproblem in Eq. (II.22) numerically, we employ a finite element method (FEM) instead of the conceptually (and programmatically) more simple finite difference method (FDM) for, mainly[2], one reason: With a FEM we are completely free in choosing the mesh (the position of the discretisation points) for the radial domain, while keeping the associated finite dimensional matrix problem real and symmetric. This is important because the problem we face presents itself with drastically different energy scales, introduced by the diverging potential: the eigenfunctions $f_{n,l,s,j}$ oscillate quickly for small arguments and slowly for large ones, where the particle is almost free. Thus, a mesh tailored to this behaviour leads to a significant reduction of computational resources (the fast oscillation can be efficiently resolved without introducing 'too many' points in the slowly oscillating regime).

The first step in setting up the FEM is to agree on a finite closed interval $I = [0, t_{\mathrm{tmax}}]$, where $t_{\max} > 0$ sets the maximal value of the domain we consider. The value $t_{\max}$ must be chosen large enough to ensure that the boundary condition $f_{n,l,s,j}(t \geq t_{\max}) = 0$ is fulfilled. This equal sign is actually never true, because the radial eigenstates asymptotically approach zero. However, if we reach zero up to machine precision, it is numerically true. Of course, one must ensure that the finite set of solutions that shall be computed has support within this interval. The mesh we define as a family $(t_k)_{k=1,\dots,N-1}$ in I with the properties $t_0 = 0$, $t_{N-1} = t_{\max}$, and $t_0 < t_1 < \cdots < t_{N-2} < t_{N-1}$, where we say $N \geq 3$. Of course, typical values are $N \sim 10^5$ - 10^7. Next, we define the derived family of step sizes $(h_k)_{k=1,\dots,N-1}$, $h_k = t_k - t_{k-1}$. The elements $(\phi_k)_{k=0,\dots,N-1}$, $\phi_k : I \to [0,1]$, we define as

$$\phi_k(t) = \begin{cases} (t - t_{k-1})/(t_k - t_{k-1}), & t \in (t_{k-1}, t_k) \\ (t_{k+1} - t)/(t_{k+1} - t_k), & t \in (t_k, t_{k+1}) \\ 0, & \text{else} \end{cases} . \tag{II.26}$$

for $k = 1, \dots, N-2$, and for $k = 0$ and $k = N-1$ the Dirichlet boundary conditions $f_{n,l,s,j}(0) = f_{n,l,s,j}(t_{\max})$ lead to the canonical choice $\phi_0 = \phi_{N-1} \equiv 0$. To obtain a matrix problem which approximates Eq. (II.22) we compute projections of it onto the elements with the real scalar product (dummy functions $\alpha, \beta : I \to \mathbb{R}$)

$$B(\alpha, \beta) = \int_I \mathrm{d}t\, \alpha(t)\beta(t) . \tag{II.27}$$

To simplify the notation we fix the quantum numbers n, l, s, and j and say $f \equiv f_{n,l,s,j}$. Moreover, we collect all potential contributions into the new potential v, defined as

$$v(t) = \frac{l(l+1)}{2t^2} + C\frac{V(a_{\mathrm{B}}t; \hbar^2 l[l+1], \hbar^2 s[s+1], \hbar^2 j[j+1])}{E_{\mathrm{H}}} . \tag{II.28}$$

Finally, we say $\lambda = \bar{E}_{n,l,s,j}$, such that Eq. (II.22) simply becomes

$$-\frac{1}{2}f''(t) + v(t)f(t) = \lambda f(t) . \tag{II.29}$$

2 It also helps to save some disk space, because one must store much less data to represent the eigenfunctions $f_{n,l,s,j}$. Typically, disk space is not a relevant factor these days, but we see no need to be unnecessarily wasteful with it either.

Before we project this equation onto the elements, we approximately expand f in them:

$$f = \sum_{k=1}^{N-2} f_k \phi_k \,. \tag{II.30}$$

This expansion automatically fulfils the Dirichlet boundary conditions $f(0) = f(t_{\max}) = 0$. Projection, i.e., applying the map $B(\phi_k, \cdot)$ for all $k = 1, \dots, N-2$, yields a matrix eigenproblem who's solutions converge to solutions of Eq. (II.22) as N tends to infinity:

$$(K + V)\boldsymbol{f} = \lambda M \boldsymbol{f} \,. \tag{II.31}$$

In this expression $M_{kk'} = B(\phi_k, \phi_{k'})$ is the so-called mass matrix, $K_{kk'} = B(\phi'_k, \phi'_{k'})/2$ the so-called stiffness matrix[3] $V_{kk'} = B(\phi_k, v\phi_{k'})$ the potential energy contribution. Each matrix is $(N-2)$-dimensional, since $k, k' = 1, \dots, N-2$, and M and K can be computed analytically; in particular

$$M = \frac{1}{6}\begin{bmatrix} 2(h_1+h_2) & h_2 & & & \\ h_2 & 2(h_2+h_3) & h_3 & & \\ & h_3 & 2(h_3+h_4) & h_4 & \\ & & \ddots & \ddots & h_{N-2} \\ & & & h_{N-2} & 2(h_{N-2}+h_{N-1}) \end{bmatrix} \tag{II.32}$$

and

$$K = \frac{1}{2}\begin{bmatrix} \frac{1}{h_1}+\frac{1}{h_2} & -\frac{1}{h_2} & & & \\ -\frac{1}{h_2} & \frac{1}{h_2}+\frac{1}{h_3} & -\frac{1}{h_3} & & \\ & -\frac{1}{h_3} & \frac{1}{h_3}+\frac{1}{h_4} & -\frac{1}{h_4} & \\ & & \ddots & \ddots & -\frac{1}{h_{N-2}} \\ & & & -\frac{1}{h_{N-2}} & \frac{1}{h_{N-2}}+\frac{1}{h_{N-1}} \end{bmatrix}. \tag{II.33}$$

Note that both of these matrices are symmetric. The potential matrix, generally, must be computed numerically. However, here we choose the simple approximation

$$B(\phi_k, v\phi_{k'}) \approx \frac{v(t_k) + v(t_{k'})}{2} M_{kk'} \,, \tag{II.34}$$

which is justified if the step size is small enough. Note that this particular choice of approximations keeps the matrix symmetric, such that $K + V$ is symmetric. Furthermore, M represents a scalar product and is thus positive definite. This means that the matrix eigenproblem in Eq. (II.31) can be efficiently solved with the Lanczos algorithm.

Next, we express the normalisation condition in Eq. (II.25) in terms of our elements and the vector $\boldsymbol{f}$. To this end we calculate

$$\int_0^\infty \mathrm{d}t\, f^2(t) \approx \int_I \mathrm{d}t\, f^2(t) = \sum_{k=1}^{N-2}\sum_{k'=1}^{N-2} f_k f_{k'} B(\phi_k, \phi_{k'}) = \boldsymbol{f}^{\mathrm{T}} M \boldsymbol{f} = \|\boldsymbol{f}\|_M^2 \overset{!}{=} 1 \,, \tag{II.35}$$

3 Note that because the elements obey the Dirichlet boundary conditions $\phi_k(0) = \phi_k(t_{\max}) = 0$, $k = 1, \dots, N-2$, integrating by parts allows to write $B(\phi''_k, \phi_{k'}) = -B(\phi'_k, \phi'_{k'})$.

i.e., the length of $\boldsymbol{f}$ must be unity in the by the scalar product M induced norm $\|\cdot\|_M$.

Finally, we compute radial matrix elements. To this end, we first calculate for $p \in \mathbb{R}$ such that the integrals exist,

$$\begin{aligned}\frac{r_p^{n,l,s,j;n',l',s',j'}}{a_\mathrm{B}^p} &\equiv \frac{1}{a_\mathrm{B}^p}\langle R_{n,l,s,j}, \hat{r}^p R_{n',l',s',j'}\rangle = \int_0^\infty \mathrm{d}r\, r^2 R_{n,l,s,j}(r)\frac{r^p}{a_\mathrm{B}^p}R_{n',l',s',j'}(r) \\ &= \int_0^\infty (\mathrm{d}t\, a_\mathrm{B})(a_\mathrm{B}t)^2 f_{n,l,s,j}(t)\frac{(a_\mathrm{B}t)^p}{a_\mathrm{B}^p}\frac{1}{a_\mathrm{B}^3}\frac{f_{n',l',s',j'}(t)}{t^2} = B(f_{n,l,s,j}, \mathrm{id}^p f_{n',l',s',j'})\,.\end{aligned} \tag{II.36}$$

Here, 'id' denotes the identity on the non-negative real numbers. Next, we approximate this result with solutions to the matrix eigenproblem in Eq. (II.31), i.e., we represent $f_{n,l,s,j}$ and $f_{n',l',s',j'}$ approximately with the vectors $\boldsymbol{f}_{n,l,s,j}$ and $\boldsymbol{f}_{n',l',s',j'}$, respectively [see Eq. (II.30)]. This allows to write

$$B(f_{n,l,s,j}, \mathrm{id}^p f_{n',l',s',j'}) \approx \boldsymbol{f}_{n,l,s,j}^\mathrm{T} R_p \boldsymbol{f}_{n',l',s',j'}\,, \tag{II.37}$$

where the components of the real symmetric matrix R_p are defined as

$$(R_p)_{k,k'} = B(\phi_k, \mathrm{id}^p\phi_{k'}) = \begin{cases} B(\phi_k, \mathrm{id}^p\phi_{k+1}), & k' = k+1 \text{ or } k = k'+1\ (0 < k, k' < N-2) \\ B(\phi_k, \mathrm{id}^p\phi_k), & k' = k\ (0 < k, k' \leq N-2) \\ 0, & \text{else} \end{cases}\,. \tag{II.38}$$

For the diagonal matrix elements we find

$$B(\phi_k, \mathrm{id}^p\phi_k) = \begin{cases} \frac{1}{(t_k-t_{k-1})^2}\left[\frac{t_k^2-t_{k-1}^2}{2} - 2t_{k-1}(t_k-t_{k-1}) + t_{k-1}^2\ln\left(\frac{t_k}{t_{k-1}}\right)\right] \\ +\frac{1}{(t_{k+1}-t_k)^2}\left[\frac{t_{k+1}^2-t_k^2}{2} - 2t_{k+1}(t_{k+1}-t_k) + t_{k+1}^2\ln\left(\frac{t_{k+1}}{t_k}\right)\right], & p = -1 \\ \frac{1}{(t_k-t_{k-1})^2}\left[t_k - t_{k-1} - 2t_{k-1}\ln\left(\frac{t_k}{t_{k-1}}\right) + t_{k-1}^2\left(\frac{1}{t_k} - \frac{1}{t_{k-1}}\right)\right] \\ +\frac{1}{(t_{k+1}-t_k)^2}\left[t_{k+1} - t_k - 2t_{k+1}\ln\left(\frac{t_{k+1}}{t_k}\right) + t_{k+1}^2\left(\frac{1}{t_{k+1}} - \frac{1}{t_k}\right)\right], & p = -2 \\ \frac{1}{(t_k-t_{k-1})^2}\left[\ln\left(\frac{t_k}{t_{k-1}}\right) - 2t_{k-1}\left(\frac{1}{t_k} - \frac{1}{t_{k-1}}\right) + t_{k-1}^2\left(\frac{1}{t_k^2} - \frac{1}{t_{k-1}^2}\right)\right] \\ +\frac{1}{(t_{k+1}-t_k)^2}\left[\ln\left(\frac{t_{k+1}}{t_k}\right) - 2t_{k+1}\left(\frac{1}{t_{k+1}} - \frac{1}{t_k}\right) + t_{k+1}^2\left(\frac{1}{t_{k+1}^2} - \frac{1}{t_k^2}\right)\right], & p = -3 \\ \frac{1}{(t_k-t_{k-1})^2}\left[\frac{t_k^{p+3}-t_{k-1}^{p+3}}{p+3} - 2t_{k-1}\frac{t_k^{p+2}-t_{k-1}^{p+2}}{p+2} + t_{k-1}^2\frac{t_k^{p+1}-t_{k-1}^{p+1}}{p+1}\right] \\ +\frac{1}{(t_{k+1}-t_k)^2}\left[\frac{t_{k+1}^{p+3}-t_k^{p+3}}{p+3} - 2t_{k+1}\frac{t_{k+1}^{p+2}-t_k^{p+2}}{p+2} + t_{k+1}^2\frac{t_{k+1}^{p+1}-t_k^{p+1}}{p+1}\right], & \text{else} \end{cases}\,. \tag{II.39}$$

We emphasise here that for $k = 1$ and p negative enough, the above expression is not well-defined. This does not mean the matrix element does not exist, because the functions $f_{n,l,s,j}$ and $f_{n',l',s',j'}$ tend to zero as $t \to 0$. If the radial matrix element exists, must therefore be determined before computing it, e.g., by considering the form of the radial equation for $t \ll 1$, and thereby

inferring the functional behaviour of the radial eigenstates close to zero. When implementing the radial matrix elements in a computer program, one may allow arbitrary (at least as long as no numerical limits are reached) p by simply setting $(R_p)_{1,1} = 0$ (we will see below that the off-diagonal components are always well-defined). This simplifying 'error' is justified by the fact that the step $h_1 = t_1 - t_0$ should be chosen small ($\sim 10^{-6}$) in order to obtain sensible results for the eigenenergies and eigenstates, as well as the fact that the eigenfunctions $f_{n,l,s,j}$ vanish at $t = 0$ fast enough (if the integral exists mathematically). Hence, the error one makes will be negligible, because, from a physical standpoint, it is clear that the radial eigenstates do not change much on the length scale $\sim 10^{-6} a_{\mathrm{B}}$.

The off-diagonal matrix elements of R_p read

$$B(\phi_k, \mathrm{id}^p \phi_{k+1}) = \begin{cases} \frac{t_{k+1}^2 - 2t_{k+1} \ln\left(\frac{t_{k+1}}{t_k}\right) t_k - t_k^2}{2(t_{k+1}-t_k)^2}, & p = -1 \\ \frac{(t_{k+1}+t_k) \ln\left(\frac{t_{k+1}}{t_k}\right) - 2(t_{k+1}-t_k)}{(t_{k+1}-t_k)^2}, & p = -2 \\ \frac{\frac{t_{k+1}}{t_k}\left(1 - \frac{t_k^2}{t_{k+1}^2}\right) - 2\ln\left(\frac{t_{k+1}}{t_k}\right)}{2(t_{k+1}-t_k)^2}, & p = -3 \\ \frac{1}{(t_{k+1}-t_k)^2}\left[\frac{t_{k+1} t_k^{p+2} - t_k t_{k+1}^{p+2}}{(p+1)(p+2)} + \frac{t_{k+1}^{p+3} - t_k^{p+3}}{(p+2)(p+3)}\right], & \text{else} \end{cases} . \tag{II.40}$$

With this result it is straight-forward to construct the matrix R_p, that is, it is straight-forward to compute arbitrary radial matrix elements.

III. Manual

Reference manual for the `AlkCalc` library functions. Prototypes for the following functions are defined in the header 'interface.d/alkcalc.h'.

FUNCTION

`alkcalc_Enlsj`

DESCRIPTION

Eigenenergies in units of Hartree. The requested eigenenergy is read from the corresponding data file in the directory 'data.d' and returned.

ARGUMENTS

`char *species`

String to specify atom/ion species, e.g., 1H for Hydrogen, or 88SR+ for the $^{88}Sr^{+}$ ion.

`int32_t n`

Principal quantum number.

`int32_t l`

Orbital angular momentum quantum number $l = 1, 2, \ldots, n-1$.

`double s`

Electron spin quantum number. Not an argument here, since $s = 1/2$ is always assumed.

`double j`

Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$. Here, the following convention is applied: For $j \in [k, k+1)$, $k = 0, 1, 2, \ldots$, the library internally uses the (half-integer) value $(2k+1)/2$. For instance, this means that for $j = 0.49999$ and $j = 0.50001$ the software internally uses $j = 1/2$ in both cases.

RETURN

`double E`

E is the eigenenergy specified by the quantum numbers n, l, $s = 1/2$, and j.

FUNCTION

`alkcalc_fnlsj`

DESCRIPTION

Numerical equivalent to $f_{n,l,s,j}$ in Eq. (II.21). The data is read from the file containing

the requested state. This data is stored at a user specified location. The location where `AlkCalc` will search for the data can be set in the file 'interface.d/alkcalc.h'. The result is owned by the caller and should be freed with `alkcalc_state_free`.

ARGUMENTS

`char result`

Character specifying if the discretisation data `t` and `h` is returned (see RETURN down below). If the argument `result` is equal to `f` (full), the full result is returned, and if `result` is equal to `p` (partial), `t=h=NULL`.

`char *species`

String to specify atom/ion species, e.g., `1H` for Hydrogen, or `88SR+` for the $^{88}Sr^{+}$ ion.

`int32_t n`

Principal quantum number.

`int32_t l`

Orbital angular momentum quantum number $l = 1, 2, \ldots, n-1$.

`double s`

Electron spin quantum number. Not an argument here, since $s = 1/2$ is always assumed.

`double j`

Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$. Here, the following convention is applied: For $j \in [k, k+1)$, $k = 0, 1, 2, \ldots$, the library internally uses the (half-integer) value $(2k+1)/2$. For instance, this means that for $j = 0.49999$ and $j = 0.50001$ the software internally uses $j = 1/2$ in both cases.

RETURN

`alkcalc_state *f`

`f` contains eight members: `N`, `dim`, `t`, `h`, and `fnlsj`.
`int32_t N`: Number of discretisation points N in the mesh, i.e., if $N = 10$, the interval $[0, t_{\max}]$ is split into $N - 1 = 10 - 1 = 9$ intervals.
`int32_t n`: Principal quantum number.
`int32_t l`: Orbital angular momentum quantum number.
`double j`: Total angular momentum quantum number.
`int32_t dim`: Dimension of eigenproblem, i.e., $N - 2$, which is the number of elements appearing in the expansion in Eq. (II.30).
`double *t`: Array of length N, containing the mesh $(t_k)_{k=0,\ldots,N-1}$.
`double *h`: Array of length $N-1$, containing the family of step sizes, $(h_k)_{k=1,\ldots,N-1}$.
`double *fnlsj`: Array of length $N - 2$ containing the requested eigenvector $\boldsymbol{f}$. The components of $\boldsymbol{f}$ are the coefficients f_k $(k = 1, \ldots, N-2)$ defining the expansion in Eq. (II.30).

FUNCTION

`alkcalc_rp`

DESCRIPTION

Radial matrix elements with arbitrary power $p \in \mathbb{R}$ [see Eq. (II.36)]. Note that for too large and too negative p the result's numerical error increases due to the finite precision used to compute differences of large numbers (catastrophic cancellation). To check for validity, use ^{1}H (Hydrogen atom) and ^{2}He^{+} (Helium ion) as a benchmark for atoms and ions, respectively. It is the user's responsibility to ensure that the integral associated to the desired matrix element exists. In case is does not, the returned values is wrong and no error is thrown (see also the discussion on p. 7).

ARGUMENTS

`char *species`

String to specify atom/ion species, e.g., 1H for Hydrogen, or 88SR+ for the ^{88}Sr^{+} ion.

`int32_t nb`

Principal quantum number of bra.

`int32_t lb`

Orbital angular momentum quantum number $l = 1, 2, \ldots, n-1$ of bra.

`double sb`

Electron spin quantum number of bra. Not an argument here, since $s = 1/2$ is always assumed.

`double jb`

Total angular momentum quantum number of bra, j. Here, the following convention is applied: For $j \in [k, k+1)$, $k = 0, 1, 2, \ldots$, the library internally uses the (half-integer) value $(2k+1)/2$. For instance, this means that for $j = 0.49999$ and $j = 0.50001$ the software internally uses $j = 1/2$ in both cases.

`double p`

Power of radius operator; cf. Eq. (II.36). For instance, $p = 0$ yields the scalar product of the states. For $p = 1$ and $p = 2$, respectively, dipole and quadrupole matrix elements are computed. Generally, $p \in \mathbb{R}$.

`int32_t nk`

Principal quantum number of ket.

`int32_t lk`

Orbital angular momentum quantum number $l = 1, 2, \ldots, n-1$ of ket.

`double sk`

Electron spin quantum number of ket. Not an argument here, since $s = 1/2$ is always assumed.

`double jk`

Total angular momentum quantum number of ket, j. Here, the convention for `jb` (see above) is adopted.

RETURN

`double rp`

`rp` is the matrix element $r_p^{n,l,s,j;n',l',s',j'}$ in Eq. (II.36) in units of a_{B}^p, where a_{B} is Bohr's radius. Quantum numbers without(with) a prime correspond to the quantum numbers of the bra(ket).

FUNCTION

`alkcalc_cj1m1j2m2jmj`

DESCRIPTION

Clebsch-Gordan coefficients,

$$C_{j,m_j}^{j_1,m_1,j_2,m_2} = \langle j_1, m_1; j_2, m_2 | j_1, j_2, j, m_j \rangle \ , \tag{III.1}$$

where j is the total angular momentum composed of the two individual ones j_1 and j_2. To fix the phase uniquely, we adopt the Condon-Shortley sign convention, that is, the Clebsch-Gordan coefficients with $m_j = j_1 + j_2$ are positive. In particular, for the case $j_1 = l$ and $j_2 = s$,

$$C_{j,m_j}^{l,m_l,s,m_s} = \langle Y_{l,m_l} \otimes \boldsymbol{\chi}_{s,m_s}, \boldsymbol{\Phi}_{l,s,j,m_j} \rangle \, , \tag{III.2}$$

with the states $Y_{l,m_l} \otimes \boldsymbol{\chi}_{s,m_s}$ of the uncoupled basis and the coupled basis states $\boldsymbol{\Phi}_{l,s,j,m}$; see Eqs. (II.12, II.16), and Eq. (II.17), respectively. The return type (see RETURN down below) allows to extract the requested coefficient symbolically as well as numerically. The function returns zero in case the arguments do not make sense, e.g., if $j_1 < m_1$; or if j_1 is half-integer and m_1 is integer; or if j does not lay between (including bounds) $|j_1 - j_2|$ and $|j_1 + j_2|$. The arguments to this function are defined as double precision floating point numbers, in order to represent half-integer values. For numerical stability and definiteness the arguments (positive as well as negative) are read with the following convention: for $a = j_1, m_1, j_2, m_2, j, m_j$ the value $a = 1/2\lfloor 2a + 1/2 \rfloor$ is used, with $\lfloor x \rfloor = \mathrm{argmin}\{|x - z| \, : \, z \in \mathbb{Z}, z \leq x\}$. This might seem somewhat convoluted, but simply means that for $-0.25 \leq a < 0.25$ we take $a = 0$; for $0.25 \leq a < 0.75$ we take $a = 1/2$; for $0.75 \leq a < 1.25$ we take $a = 1$; and so on (in both positive and negative direction). This makes sure that the finite precision of double precision floating point numbers does not play a role; i.e., it does not matter if one puts $a = 0.5$ or $a = 0.500001$ or $a = 0.499998$ as an argument — $a = 1/2$ is used internally in any case.

ARGUMENTS

`double j1`

Angular momentum quantum number of first factor.

`double m1`
Projection quantum number of first factor.

`double j2`
Angular momentum quantum number of second factor.

`double m2`
Projection quantum number of second factor.

`double j`
Total angular momentum quantum number.

`double mj`
Projection quantum number of total angular momentum.

RETURN

`alkcalc_cg c`
`c` contains three members: `sign`, `numerator`, and `denominator`.
`int8_t sign`: Sign, σ, of the Clebsch-Gordan coefficient.
`int64_t numerator`: Numerator, ν, of the Clebsch-Gordan coefficient.
`int64_t denominator`: Denominator, μ, of the Clebsch-Gordan coefficient.
The requested Clebsch-Gordan coefficient is given by $\sigma\sqrt{\nu/\mu}$. Euklid's algrithm is employed to reduce ν and μ. This representation allows to obtain Clebsch-Gordan coefficients symbolically as well as numerically.

FUNCTION

`alkcalc_YlmlXsms`

DESCRIPTION

Angular factor of the full eigenstate in the uncoupled basis, i.e., the two-component spinor $Y_{l,m_l}(\vartheta,\varphi)\boldsymbol{\chi}_{s,ms}$. The input quantum numbers are checked for inconsistencies and an error is thrown in case one is detected. The angles ϑ and φ have the ranges $[0,\pi]$ and $[0,2\pi]$, respectively. This condition is checked up to floating point machine precision (type `double`) and an error will be thrown in case either condition is not obeyed.

ARGUMENTS

`int32_t l`
Orbital angular momentum quantum number $l = 1, 2, \ldots, n-1$.

`int32_t ml`
Projection quantum number of orbital angular momentum.

`double s`

Electron spin quantum number. Not an argument here, since $s = 1/2$ is always assumed.

`double ms`

Projection quantum number of electron spin, m_s. For any $m_s < 0$ the value $m_s = -1/2$ is used internally, and for $m_s \geq 0$ the value $m_s = 1/2$.

`double theta`

Polar angle (Zenitwinkel), ϑ, in range $[0, \pi]$.

`double phi`

Azimuthal angle (Azimut), φ, in range $[0, 2\pi]$.

RETURN

`alkcalc_spinor s`

`s` has two components and represents a two-component spinor: `u` is the spin-up ($m_s = 1/2$) component and `d` is the spin-down ($m_s = -1/2$) component.

FUNCTION

`alkcalc_Philsjmj`

DESCRIPTION

Angular factor of the full eigenstate in the coupled (fine-structure) basis, i.e., the two-component spinor $\Phi_{l,s,j,mj}(\vartheta, \varphi)$ [see Eq. (II.17)]. The input quantum numbers are checked for inconsistencies and an error is thrown in case one is detected. The angles ϑ and φ have the ranges $[0, \pi]$ and $[0, 2\pi]$, respectively. This condition is checked up to floating point machine precision (type `double`) and an error will be thrown in case either condition is not obeyed. To compute the resulting two-component spinor Clebsch-Gordon coefficients are needed, for which the sign convention detailed in Eq. (III.2) is employed.

ARGUMENTS

`int32_t l`

Orbital angular momentum quantum number $l = 1, 2, \ldots, n-1$.

`double s`

Electron spin quantum number. Not an argument here, since $s = 1/2$ is always assumed.

`double j`

Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$. Here, the following convention is applied: For $j \in [k, k+1)$, $k = 0, 1, 2, \ldots$, the library internally uses the (half-integer) value $(2k+1)/2$. For instance, this means that for $j = 0.49999$ and $j = 0.50001$ the software internally uses $j = 1/2$ in both cases.

`double mj`

Projection quantum number of total angular momentum, m_j. Regarding the floating point representation of half-integers, the same convention as for the total angular momentum quantum number j (see above) is employed.

`double theta`

Polar angle (Zenitwinkel), ϑ, in range $[0, \pi]$.

`double phi`

Azimuthal angle (Azimut), φ, in range $[0, 2\pi]$.

RETURN

`alkcalc_spinor s`

`s` has two components and represents a two-component spinor: `u` is the spin-up ($m_s = 1/2$) component and `d` is the spin-down ($m_s = -1/2$) component.

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